

DharmaTej Vanamamaly

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CAREER OBJECTIVE

Final-year B.Tech student in Computer Science Engineering (AIML) at SR University, Warangal, with hands-on experience in machine learning, deep learning, and full stack web development. Passionate about building intelligent, data-driven systems and deploying scalable applications.

EXPERIENCE

AI-ML Virtual Intern

2024 – Present

EduSkills / Virtual Internship

- Built end-to-end ML pipelines from scratch, performing data cleaning, feature engineering, and model development using multiple algorithms to select the best-performing solution.
- Optimized model efficiency and predictive accuracy through systematic algorithm benchmarking, hyperparameter tuning, and regularization techniques.
- Designed reproducible data pipelines ensuring consistency across experiments, reinforcing production-ready ML development practices.

PROJECTS

AI Playlist Generator | *Deep Q-Learning, TensorFlow, Python, Streamlit, Spotify API*

- Engineered a reinforcement learning-based music recommendation system using Deep Q-Networks (DQN) to generate dynamic, personalized playlists based on user preference and listening patterns.
- Designed a mood-aware playlist sequencing algorithm to improve contextual relevance by optimizing song transitions, applying reward-based learning to maximize user engagement.
- Integrated Spotify API for real-time data ingestion and deployed the solution via Streamlit, demonstrating end-to-end problem solving from data ingestion to deployment.

Household Electricity Consumption Prediction | *Python, sikit-learn, ML Algorithms*

- Built and benchmarked multiple supervised ML models (Linear Regression, SVM, KNN, Ridge, Lasso) on real-world energy datasets to predict household electricity consumption.
- Applied systematic data preprocessing and feature engineering pipelines, improving predictive accuracy through hyperparameter tuning and regularization techniques.
- Conducted algorithm comparison analysis to identify optimal model, reinforcing structured problem-solving and data-driven decision-making.

TECHNICAL SKILLS

Languages: Python, Java

Data Structures & Algorithms: Arrays, Linked Lists, Trees, Graphs, Sorting, Searching, Dynamic Programming, Recursion

AI / ML / GenAI: Deep Learning, Reinforcement Learning, TensorFlow, Keras, Regression Models, Classification Algorithms, Prompt Engineering, LLM Fundamentals

Web & Databases: HTML, MySQL, MongoDB

CS Fundamentals: OOP, Operating Systems, DBMS

Cloud & Tools: Microsoft Azure (AI Fundamentals Certified), Git, Streamlit

EDUCATION

SR University

B.Tech in Computer Science Engineering (AI & ML) — CGPA: 8.55 / 10.0

Warangal, Telangana

2023 – 2027

SR Junior College (TSBIE Board)

Senior Secondary (Class 12) — 81.9%

Telangana

2021 – 2023

CERTIFICATIONS

Microsoft Certified: Azure AI Fundamentals

February 2025

Microsoft

Deep Learning for Developers

April 2025

Infosys Springboard

Python Full Stack Developer Virtual Internship

October. – December. 2024

EduSkills

ACHIEVEMENTS

- **Vice-Chancellor's Certificate of Appreciation** — Awarded by SR University for outstanding academic excellence (Top performer, CGPA: 8.55).
- **National Level Environmental Sustainability Exhibition (ESE-2025)** — Selected as a participant at SRM University, demonstrating initiative beyond academics.